

Xavier Kervyn

Max Planck Institute for Physics, Boltzmannstraße 8, 85748 Garching bei München, Germany
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CURRENT RESEARCH INTERESTS

★ String amplitudes & dualities, holography, positive geometry, algebraic geometry, resurgence theory

ACADEMIC EMPLOYMENT

Max Planck Institute for Physics Garching bei München, Munich, Germany	10/2025
Doctoral Researcher, MPP QFT + String Theory Groups	–present
– International Max Planck Research School (IMPRS) PhD candidate	
– Advised by Stephan Stieberger (string theory) and Johannes Henn (quantum field theory)	
Max Planck Institute for Physics Garching bei München, Munich, Germany & remote	09/2024
Graduate Researcher, MPP QFT + String Theory Groups	09/2025
– Master’s thesis on string amplitudes and flat-space holography	
– Advised by Stephan Stieberger (string theory)	

EDUCATION

Ludwig-Maximilians-Universität München Munich, Germany	10/2025
PhD in Theoretical Physics	–present
– Advanced coursework in mathematics: algebraic geometry, resurgence theory	
Ludwig-Maximilians- & Technische Universität München Munich, Germany	10/2023
Master of Science in Theoretical and Mathematical Physics (TMP, with Distinction)	09/2025
– Thesis: <i>Towards a String Worldsheet Description of Celestial Holography</i> (1.0, High Distinction, 1 paper)	
– Awarded 2 scholarships. Research at Niels Bohr Institute & Max Planck Institute for Physics	
Peterhouse, University of Cambridge Cambridge, United Kingdom	10/2022
Master of Advanced Study in Applied Mathematics (Part III, with First Class Honours)	06/2023
– Part III essay: <i>BMS Symmetries of Gravitational Scattering</i> (86%, Distinction, published as review)	
– Awarded 3 scholarships and 3 prizes. Summer research at Università degli Studi di Padova	
Ecole Polytechnique Fédérale de Lausanne Lausanne, Switzerland	09/2019
Bachelor of Science in Physics	08/2022
– Awarded 2 fellowships and 1 scholarship. Research at CERN & ETH Zürich	

RESEARCH STAYS & LONG-TERM ACADEMIC VISITS

Niels Bohr Institute Copenhagen, Denmark	09/2024
Visiting Student, Theoretical High-Energy Physics	01/2025
– Study project on the superconformal bootstrap of gauge theories with defects	
– Supervised by Charlotte Kristjansen and Adam Chalabi	
Università degli Studi di Padova Padova, Italy	07/2023
Visitor, Theoretical Physics	
– Research on integrability in superconformal field theories and AdS/CFT	
– Supervised by Alessandro Sfondrini and Davide Polvara	

CERN CMS collaboration Meyrin, Switzerland	06/2022
Undergraduate Researcher, ETH Zürich High-Energy Physics group	07/2022
– Wrote a Python package for the new high-luminosity CMS electrocalorimeter, detector testing	
ETH Zürich Zürich, Switzerland	09/2021
Undergraduate Researcher, Exoplanets and Habitability group	12/2021
– Experimental concept of the LIFE mission, modelled imperfect nulling in the interferometer	

SELECTED AWARDS, GRANTS & SCHOLARSHIPS

Bourse d'Études – <i>Fondation du Domaine de Villette</i> // CHF 5'000	10/2024
Erasmus+ Scholarship – <i>Ludwig-Maximilians-Universität München</i> // €3'000	05/2024
College Prize in Applied Mathematics – <i>Peterhouse, University of Cambridge</i> // £300	12/2023
Retrospective Scholarship in Mathematics – <i>Peterhouse, University of Cambridge</i> // £125	12/2023
Bourse d'Études – <i>Fondation du Domaine de Villette</i> // CHF 5'000	06/2023
Greta Burkill Fund Award – <i>Peterhouse, University of Cambridge</i> // £300	03/2023
Bruckmann Fund Award – <i>Peterhouse, University of Cambridge</i> // £300	03/2023
Annual Scholarship – <i>Swiss Study Foundation</i> // CHF 20'000	11/2022
Excellence Scholarship – <i>Colbianco Stiftung</i> // CHF 2'000	08/2022
Fellowship – <i>Swiss Study Foundation</i> & <i>e-fellows.net</i>	05/2022
Swiss Mobility Program Grant – <i>Ecole Polytechnique Fédérale de Lausanne</i> // CHF 1'500	09/2021
Baccalaureate Merit Award – <i>Région Provence-Alpes Côte d'Azur</i> // €400	07/2019

PUBLICATIONS & PREPRINTS ([Inspire HEP](#); *:under review) (author ordering is alphabetical in hep-th)

Articles:

3. X. Kervyn and S. Stieberger, *Resurgence of high-energy string amplitudes*, [2604.07444](#).
2. X. Kervyn and S. Stieberger, *High-energy string theory and the celestial sphere*, *Journal of High Energy Physics* **09** (2025) 044 [[2504.13738](#)]. (based on my master thesis at LMU/TUM)
1. X. Kervyn, D. Polvara and A. Sfondrini, *Thermodynamics of integrable $\mathcal{N} = 2$ theories, squared*, *Journal of High Energy Physics* **09** (2025) 018 [[2502.10356](#)].

Reviews:

1. X. Kervyn, *BMS symmetries of gravitational scattering*, *Nuclear Physics B* **1017** (2025) 116948 [[2308.12979](#)]. (based on my Part III essay at Cambridge)

PARTICIPATION TO CONFERENCES (*:scheduled, P:poster, T:invited talk)

11. Fifty Years of BRST, A. Sommerfeld Center for Theoretical Physics, Munich, Germany [link]	03/2026
10. Symbology @15, MPI MPP, Munich, Germany [link]	12/2025
9. Feynman Calculus and its Applications to Gravity and Particle Physics [link , <i>online</i>]	11/2025
8. DFG meeting: Modern Foundations of Scattering Amplitudes, BCTP, Bonn, Germany [link]	11/2025
7. From Good Cuts to Celestial Holography, St Anthony's College, Oxford University, UK [link]	07/2025
6. Infrared Surprises of Scattering Amplitudes, CERN, Meyrin, Switzerland [link , <i>online</i>]	05/2025
5. Strings 2024, CERN, Meyrin, Switzerland [link]	06/2024
4. Carrollian Physics and Holography, ESI, Vienna, Austria [link]	04/2024
3. Workshop [...] in Analysis and Mathematical Physics, LMU, Munich, Germany [link]	10/2023
2. Integrability, Dualities and Deformations 2023, Durham University, Durham, UK [link]	07/2023
1. Eurostrings 2023, Universidad de Oviedo, Gijon, Spain [link]	04/2023

PARTICIPATION TO POSTGRADUATE SCHOOLS AND WORKSHOPS (*:scheduled, P:poster, T:invited talk)

12. Prospects in Theoretical Physics: Celestial Holography, IAS Princeton, NJ, USA [link]	* 07/2026
11. Hidden Structures in Multi-loop Amplitudes, Mainz ITP, Germany [link, online]	04/2026
10. Universe+ School: Exploring Positive Geometry, les Houches, France [link]	03/2026
9. Single-valued Periods in Scattering Amplitudes, Mainz ITP, Germany [link, online]	01/2026
8. Geometry and Combinatorics of Scattering Amplitudes, MPI MiS, Leipzig, Germany [link]	10/2025
7. School on Asymptotic Symmetries and Flat Holography, G. Galilei Inst., Florence, Italy [link]	(P) 05/2025
6. Physics of Machine Learning & ML for physics, Nordita, Stockholm, Sweden [link, online]	01/2025
5. Frontiers in Gravity, Niels Bohr Institute, Copenhagen, Denmark [link]	09/2024
4. PhD school: Towards Gravity, Nordita, Stockholm, Sweden [link, online]	08/2024
3. Celestial Holography Summer School, Perimeter Institute, Waterloo, Canada [link, online]	07/2024
2. Celestial 2023, University of Warsaw, Warsaw, Poland [link]	08/2023
1. Young Researchers Integrability School & Workshop, Durham University, Durham, UK [link]	07/2023

INVITED TALKS & SEMINARS

Wall-crossing structure from quantum phenomena to Feynman integrals Quantum Field Theory Journal Club, Max Planck Institute for Physics	05/2026
Resurgence of high-energy string amplitudes String Group Seminar, LMU Munich & Max Planck Institute for Physics	05/2026
Quantum Geometry Group Seminar, Technical University of Munich	05/2026
Theory Seminar, Niels Bohr Institute, Copenhagen	04/2026
Matrix theory reloaded: a BPS road to holography String Theory Lunch Seminar, LMU Munich & Max Planck Institute for Physics	11/2025
Boundary Carrollian conformal field theories and open null strings String Theory Lunch Seminar, LMU Munich & Max Planck Institute for Physics	06/2025
Spinor-helicity formalism, twistor theory and scattering amplitudes Seminar on Scattering Amplitudes, LMU Munich	05/2024
Gauging of discrete higher-form symmetries in non-Abelian Yang-Mills theory Seminar on Generalized Symmetries in QFT, LMU Munich	12/2023
Gravitational scattering and covariant phase space methods in gravity DAMTP Part III Seminar Series, University of Cambridge	03/2023
Holography and twistor methods in AdS_5 DAMTP Part III Seminar Series, University of Cambridge	12/2022

TECHNICAL SKILLS

Languages: French (native speaker); English, German (full work proficiency); Danish (basics)
IT: Object-oriented programming (C++, Python), *Mathematica*, *MATLAB*, $\text{\LaTeX}$ (GitHub+VSCode/Overleaf)

ACADEMIC REFERENCES (thesis advisory committee)

- **Dr. Stephan Stieberger**
Permanent Staff Member & Research Group Leader, String Theory, Max Planck Institute for Physics
Email: stephan.stieberger@mpp.mpg.de
- **Prof. Dr. Johannes Henn**
Director & Department Head, Quantum Field Theory, Max Planck Institute for Physics
Email: johannes.henn@mpp.mpg.de
- **Prof. Dr. Niels Obers**
Professor & Institute Research Deputy Head, Niels Bohr Institute
Email: obers@nbi.ku.dk

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